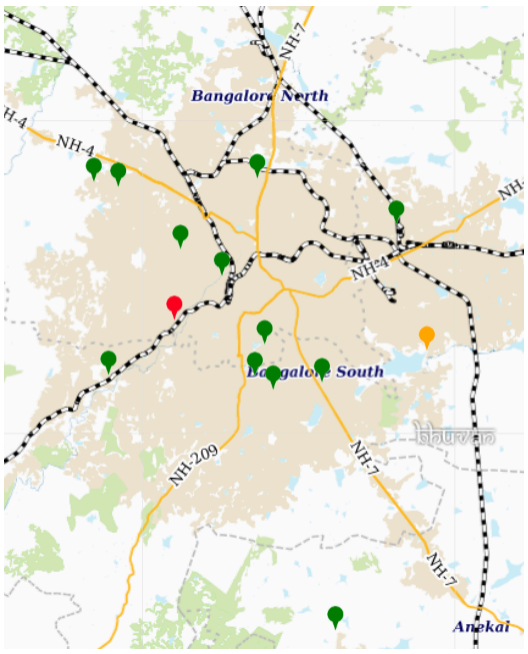
Laying a New Foundation for Clean Air in the Neighbourhoods of our Future

Air pollution is increasingly a nationwide crisis. It is not just a Delhi problem or a North India problem. Cities like Bengaluru and Mumbai also face increasing air pollution. Air quality data is critical to measuring and managing the problem all across India. The Central and State Pollution Control Boards have made progress over the past decade in installing reference-grade continuous ambient air quality monitors (CAAQMS) in many cities (562 stations as of writing), and these data are available through the SAMEER app and the CPCB website.

However, the densification of our urban areas and dynamic growth patterns have resulted in significant areas of our cities not being monitored, especially at the neighbourhood level (~every square kilometer).



For example, the figure on the left shows the current monitoring status for Bengaluru (source: CPCB).

While there is increasing recognition of the airshed-wide nature of urban air pollution, a significant fraction of urban $PM_{2.5}$ and PM_{10} air pollution can still be attributed to local sources such as traffic, open waste burning (because of inadequate solid waste management), construction (buildings, metros, road repairs), residential and roadside solid biofuel use, and resuspended road dust (especially from unpaved roads). For example, recent preliminary research at CSTEP estimates that almost half of Bengaluru's $PM_{2.5}$ comes from local sources. A pilot sensor network deployed last year across Bengaluru identified hyperlocal hot spots where $PM_{2.5}$ can be as much as 30% higher than the city-wide average; hyperlocal contributions to PM_{10} can be even larger.

City-wide dense monitoring networks enabling neighbourhood-scale air quality monitoring can enable clean air action by varied stakeholders:

1. **Urban local bodies (ULBs)** can use neighbourhood-level data to take targeted action at air pollution hot spots identified through a measurement-based approach rather than based on emission inventory-based estimates (which are based on approximated activity data and assumed emission factors). Such actions can include ensuring (1) efficient disposal of construction and demolition (C&D) waste; (2) clean construction practices are followed; (3) planning and monitoring interventions like low-emission zones (LEZs).

2. **Citizen groups** and **corporations** can use hyperlocal data to drive clean air action in their neighbourhoods, such as better solid waste management, curbing open waste burning, encouraging clean construction practices, and helping with disposal of C&D waste, so that they (and especially their children) breathe cleaner air.

3. **To support indicative health alerts** that advise vulnerable community members (seniors and children) to stay indoors (where air purifiers can be effectively used) and outdoor workers (such as traffic police) to protect themselves from air pollution by wearing a high-quality mask.

4. **Researchers** can use the data to improve air quality models and emission inventories, and make better health impact assessments.

A recent meeting convened by the ClimateRISE Alliance showed that some of these activities are already underway in cities across India, such as dust monitoring at construction sites (by the BMC, advised by CSTEP; CEEW/Earthmetry are building the consolidated dashboard) and more efficient disposal of C&D waste (led by A-PAG in Delhi). CEEW is working on the use of satellite imagery to better identify open waste burning, while Jhatka has initiated citizen-reporting of trash burning and waste dumping in Bengaluru. Research by institutions like IIT Kanpur and IIT Delhi is underway to provide hyperlocal source apportionment and better characterise urban-rural gradients, especially in North India. These efforts need to be scaled up significantly. Dense air quality sensing networks can play a foundational role in these efforts by identifying problem areas, validating scientific models, and tracking the effectiveness of implemented solutions.

However, increasing the number of CAAQM stations to ensure ubiquitous neighbourhood-scale air quality monitoring is not feasible due to their high capital cost, infrastructure needs, and requirement for ongoing maintenance by trained personnel. This is where air quality sensors, also called low-cost sensors or microsensors, can play an important role. For example, IIT Kanpur recently launched the 75-node Mumbai Air Network for Advanced Sciences (MANAS) and have shared their district-level sensor-based air quality data for the states of UP and Bihar on the Atman AQ portal. Sensors are much more portable and compact (e.g. they can be installed on a light pole or compound wall), do not need an air-conditioned chamber, and can provide indicative air quality data at <5% of the cost of a reference monitor (CAAQMS). Initial and periodic (every 6-12 months) collocations with reference monitors assure data quality. The India air quality Sensor Evaluation and Training facility (Indi-SET) at CSTEP's Bengaluru office provides such sensor collocation services; a network of Indi-SET facilities is proposed across India.

To provide transparent access to neighbourhood-scale air quality data to ULBs, citizen groups, and researchers, there is a pressing need to deploy dense networks of carefully calibrated sensors across our cities. Hence, a consortium of research institutions, air quality sensor makers, neighbourhood action groups, and philanthropic funders will be launching a ***new open data portal*** that will initially provide hyperlocal $PM_{2.5}$/PM_{10} data in the city of Bengaluru. The goal is to make more hyperlocal air quality data available so people can use participatory science to improve quality of life in their neighbourhoods. ***The new portal*** will initially collect data from more than 150 carefully-calibrated sensors hosted by residents, education institutions, and office complexes across the city. This open data portal allows all stakeholders, including government agencies, to tap into the data via APIs and make inferences that lead to clean air action. The consortium is hoping to expand the network across the country to more than 3,000 hyperlocal sensors in the next phase.